Fiscal Year:	FY 2011	**Task Last Updated:**	FY 09/08/2011
PI Name:	Bloomberg, Jacob J. Ph.D.		
Project Title:	Development of Countermeasures to Enhance Sensorimotor Adaptation		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Sensorimotor Adaptation Team		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058-3607	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Crew Health NNJ07ZSA002N
Start Date:	10/01/2008	**End Date:**	09/30/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Cohen, Helen (Baylor College of Medicine) Reschke, Millard (NASA Johnson Space Center) Mulavara, Ajitkumar (USRA) Buccello-Stout, Regina (The University of Texas Medical Branch) Peters, Brian (Wyle Laboratories)		
Grant/Contract No.:	NCC 9-58-SA01601		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	During adaptation to novel gravitational environments, sensorimotor disturbances have the potential to disrupt the ability of astronauts to perform required mission tasks. The overall goal of this project is to gain the information necessary to develop a comprehensive sensorimotor adaptability (SA) training program to facilitate rapid adaptation to novel gravitational environments. It has been previously shown that subjects trained to adapt to varied sensorimotor challenges can adapt faster to new sensory environments that they have never experienced before. This is a process known as adaptive generalization and it allows you to enhance the ability to "learn how to learn" to adapt to novel environments. By applying these motor-learning concepts for training astronauts we can enhance their ability to rapidly adapt their behavioral responses following a gravitational transition. To minimize cost and demands on crew time we have integrated SA training with existing exercise activities, namely treadmill walking. The SA training program we are developing entails manipulating the sensory conditions of treadmill exercise to systematically challenge multiple sensorimotor systems while conducting nominal exercise activities. To provide SA training we have mounted a treadmill on a six degree-of-freedom motion base to produce variation in the support surface along with variation in visual input during walking using both a projected virtual scene that produces variation in visual flow or through goggles that distort visuomotor information. Below is an overview of our research results to-date: Study 1. Aim: Determine whether sensorimotor adaptability (SA) skills learned during treadmill SA training could transfer to a novel discordant sensory environment and to quantify the extent to which any training effects would be retained. This study showed that SA training enhances locomotor adaptability and increases multi-tasking capability during adaption to a novel discordant sensory environment. Results also demonstrated that this increased adaptability can be retained up to 6 months and perhaps longer. Thus SA training improves both locomotor function and multi-tasking capabilities which are essential components required to complete critical mission tasks. Study 2. Aim: Determine whether performance in a novel discordant sensory environment can be predicted based on a subject's inherent visual dependency. This study showed that subjects with greater visual dependency have decreased locomotor stability and reduced ability to multi-task when negotiating novel sensory discordant conditions. These data indicate that visual dependency may be a "marker" for decreased ability to adapt to novel environments. Identifying preflight predictors of sensorimotor adaptability can be used to develop individualized training prescriptions that target the specific needs of each crewmember thus making the training process targeted and more efficient. Study 3. Aim 1: Examine the influence of visual dependence on a subject's SA training performance. Aim 2: Determine whether visual dependence influences a person's ability to generalize newly acquired adaptive skills in unfamiliar discordant sensory conditions. This study demonstrated that highly visually dependent individuals receive the greatest benefit from SA training. Although both visually dependent and independent subjects perform equally well by the end of their training, the visually independent subjects outperformed their highly visually dependent counterparts on a locomotor test in a novel sensory discordant environment. This suggests that visually dependent subjects may receive the greatest benefit from SA training but they are less able to generalize their skills. Study 4. Aim: Determine if SA training using treadmill walking transfers to other functional tasks including manual control. Subjects performed two operationally oriented manual control tests after locomotor SA training. Results indicate that training tasks need to contain at least some of the critical features of the criterion task to be effective. This result informs the design of future training programs ensuring that they contain sufficient locomotor and manual challenges to allow for a comprehensive training across performance modalities. Study 5. Aim: Characterize the metabolic cost of locomotor instability and potential reduction in metabolic costs due to sensorimotor adaptive training. This interdisciplinary study was conducted in collaboration with the Exercise Physiology Laboratory at NASA Johnson Space Center. To determine if SA training can reduce the metabolic cost of locomotor instability subjects walked on a treadmill for an 8-minute baseline walking period followed by 20-minutes of walking with support surface motion. Metabolic data were collected via a gas analysis facemask system along with stride frequency, and reaction time measures. At the onset of support surface motion, oxygen consumption increased. Throughout the 20-minute adaptation period, VO2 gradually decreased toward baseline levels in a manner similar to that of stride frequencies and reaction times. These data indicate that walking in discordant conditions not only compromises locomotor stability and the ability to multi-task, but comes at a quantifiable metabolic cost. Importantly, like locomotor stability and multi-tasking, metabolic cost expenditure while walking in discordant sensory conditions improved during adaptation, confirming that SA locomotor training benefits extend further than locomotor stability and multi-tasking improvements.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Sensorimotor adaptability training programs have Earthbound application in rehabilitation of patients with balance disorders, and for fall prevention training among seniors. The most important factors underlying morbidity in the older adult population are injurious falls and the restriction of activity due to falls. Approximately 25-35% of community-dwelling persons older than 65 years fall at least once a year, and approximately 40-50% of fallers experience two or more falls. The outcome is that 40% of all nursing home admissions are due to injuries due to falls. Age-related sensory changes and deterioration in ability to compensate in older adults can contribute to increase incidence of falling. We previously conducted a study to determine if balance training using variation in visual flow during treadmill exercise improves functional mobility in healthy older adults who were experiencing age-related postural instabilities (Buccello-Stout et al. 2008). The results showed that subjects who were exposed to varied visual flow during treadmill walking significantly improved their ability to negotiate an obstacle course after training compared to another elder group who only walked on a treadmill for the same amount of time. Importantly, the training benefit was retained when subjects were tested four weeks later. This study confirms that adaptability training developed for use by astronauts can also be used to improve balance and gait performance in elder subjects and points to the general applicability of this type of training in different clinical populations.

Task Progress:	1) In FY11 we completed the data collection and analysis for the third, fourth, and fifth studies in this project. 2) During this review period six presentations related to this project were completed at three international scientific meetings. These include: - 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. - 8th Symposium on the Role of the Vestibular Organs in Space, Houston, TX, April 8-10, 2011. - 82nd Annual Scientific Meeting of the Aerospace Medical Association, Anchorage, AK, May 9-12, 2011. 3) One abstract was submitted to the Major Problems of Space Life Sciences Conference, Moscow, Russia for presentation in October 2011. 4) Two manuscripts were published and two submitted for review.
Bibliography Type:	Description: (Last Updated: 11/14/2025)
Abstracts for Journals and Proceedings	Bloomberg JJ, Peters BT, Mulavara AP, Brady RA, Batson CD, Miller CA, Cohen HS. "Countermeasures to enhance sensorimotor adaptability." Major Problems of Space Life Sciences Conference, Moscow, Russia, October 20-21. 2011. Major Problems of Space Life Sciences Conference, Moscow, Russia, Abstracts, October 2011. , Oct-2011
Abstracts for Journals and Proceedings	Bloomberg JJ, Peters BT, Mulavara AP, Brady RA, Batson CD, Ploutz-Snyder RJ, Cohen HS. "Training enhances both locomotor and cognitive adaptability to a novel sensory environment." 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. 18th IAA Humans in Space Symposium Abstracts, Houston, TX, April 11-15, 2011. , Apr-2011
Abstracts for Journals and Proceedings	Bloomberg JJ, Peters BT, Mulavara AP, Brady RA, Batson CD, Ploutz-Snyder RJ, Cohen HS. "Training to facilitate adaptation to novel sensory environments." 8th Symposium on the Role of the Vestibular Organs in Space Exploration, Houston, TX, April 8-10, 2011. 8th Symposium on the Role of the Vestibular Organs in Space Exploration, Abstracts, April 2011. , Apr-2011
Abstracts for Journals and Proceedings	Brady RA, Batson CD, Peters BT, Mulavara AP, Bloomberg JJ. "Visual bias predicts gait adaptability in novel sensory discordant conditions." 8th Symposium on the Role of the Vestibular Organs in Space Exploration, Houston, TX, April 8-10, 2011. 8th Symposium on the Role of the Vestibular Organs in Space Exploration, Abstracts, April, 2011. , Apr-2011
Abstracts for Journals and Proceedings	Peters BT, Brady RA, Miller CA, Lawrence EM, Mulavara AP, Bloombert JJ. "Reduction in dynamic visual acuity reveals gaze control changes following spaceflight." 8th Symposium on the Role of the Vestibular Organs in Space Exploration, Houston, TX, April 8-10, 2011. 8th Symposium on the Role of the Vestibular Organs in Space Exploration, Abstracts, 2011. , Apr-2011
Abstracts for Journals and Proceedings	Peters BT, Brady RA, Miller CA, Mulavara AP, Cohen HS, Wood SJ, Bloomberg JJ. "Dynamic visual acuity: a functionally relevant research tool." 82nd Annual Scientific Meeting of the Aerospace Medical Association, Anchorage, AK, May 8-12, 2011. Aviation, Space, and Environmental Medicine 2011 Mar;82(3):243-4. http://www.ingentaconnect.com/content/asma/asem/2011/00000082/00000003 ;jsessionid=61tp278blca3e.alexandra , Mar-2011
Articles in Other Journals or Periodicals	Brady RA, Brian BT, Bloomberg JJ. "Walking on an oscillating treadmill: Strategies of stride-time adaptation." Ecological Psychology. In press, August 2011. , Aug-2011
Articles in Peer-reviewed Journals	Peters BT, Miller CA, Brady RA, Richards JT, Mulavara AP, Bloomberg JJ. "Dynamic visual acuity during walking after long-duration spaceflight." Aviation, Space, and Environmental Medicine. 2011 Apr;82(4):463-6. PMID: 21485405 , Apr-2011
Articles in Peer-reviewed Journals	Batson CD, Brady RA, Peters BT, Ploutz-Snyder RJ, Mulavara AP, Cohen HS, Bloomberg JJ. "Gait training improves performance in healthy adults exposed to novel sensory discordant conditions." Experimental Brain Research. 2011 Apr;209(4):515-24. Epub 2011 Feb 25. PMID: 21350808 ; http://dx.doi.org/10.1007/s00221-011-2574-6 , Apr-2011
Awards	Bloomberg J. "NASA Performance Award, July 2011." Jul-2011